SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510** **Slot A**

COURSE OUTCOME COVERED

CO4: ***Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*** **Assessment Tool Weightage: 40%**

QUESTIONS:5 **Total Marks 50** **Duration :60 Minutes Annexure A**

Data Interpretation

Code of Conduct:

I P. Akhila / 192472320 ***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.***

Signature of the student: P. Akhila

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- **Viola-Jones: fast but less accurate (78%)**
- **YOLO: moderate speed, good accuracy (92%)**
- **Deep Learning: high accuracy (95%) but slow**

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

Real-Time System Design Trade-off

Given Data

Technique	Speed	Accuracy
Viola-Jones	Fast	78%
YOLO	Moderate	92%
Deep Learning	Slow	95%

Data Interpretation

The three methods show a clear trade-off between processing speed and detection accuracy. Viola-Jones provides the fastest operation but has the lowest accuracy of 78%. YOLO provides a better balance, with 92% accuracy and moderate speed. Deep Learning gives the highest accuracy of 95%, but its processing speed is slow.

Analysis

Under strict latency constraints, the system cannot choose a method only based on maximum accuracy. A surveillance system must respond quickly to detected events, so a very slow method may cause delays. Viola-Jones satisfies the speed requirement, but the 14% accuracy gap compared with YOLO is significant. YOLO improves accuracy substantially while still maintaining moderate processing speed. Deep Learning gives only a further 3% accuracy improvement over YOLO, but with a much higher speed penalty.

Application

For a real-time surveillance system, YOLO is the most suitable choice because it provides a practical balance between detection accuracy and processing speed. Hardware acceleration or model optimization can further improve its latency if required.

Conclusion

YOLO should be selected under strict latency constraints. Although Viola-Jones is faster, its 78% accuracy may result in too many missed detections, while the additional accuracy of Deep Learning does not justify its slow operation in a strict real-time system.

2. **Robust Detection under Environmental Variations Detection accuracy across environments:**

Method Normal Occlusion Low Light

Viola-Jones 80% 60% 55%

YOLO 90% 75% 70%

Deep Learning 95% 85% 80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

Robust Detection under Environmental Variations

Given Data

Method	**Normal**	**Occlusion**	**Low Light**
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Data Interpretation

Deep Learning provides the highest accuracy in all three environments: 95% under normal conditions, 85% with occlusion, and 80% in low light. YOLO ranks second with 90%, 75%, and 70%, while Viola-Jones gives the lowest results, particularly under low-light conditions.

The average accuracy is approximately:

- Viola-Jones: 65%
- YOLO: 78.3%
- Deep Learning: 86.7%

Analysis

The data shows that all methods experience reduced accuracy when environmental conditions become difficult. However, Deep Learning has the smallest overall performance degradation compared with the other methods.

From normal to low-light conditions:

- Viola-Jones decreases by 25 percentage points.
- YOLO decreases by 20 percentage points.
- Deep Learning decreases by only 15 percentage points.

This indicates that Deep Learning is more robust to environmental variations.

Application

For applications such as outdoor surveillance, autonomous systems, or monitoring where lighting and occlusion can change, Deep Learning is the most appropriate method. If computational resources are limited, YOLO can be considered as a compromise.

Conclusion

Deep Learning provides the best overall robustness, achieving the highest accuracy in every tested condition and maintaining better performance under occlusion and low light. YOLO is a reasonable alternative when computational constraints are important.

3. **Optical Flow Reliability vs Noise**

Sensitivity Two optical flow methods show:

- **Method A: stable vectors but misses small motions**
- **Method B: detects fine motion but noisy vectors**

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

Given Data

- Method A: Stable motion vectors but misses small motions.
- Method B: Detects fine motion but produces noisy vectors.

Data Interpretation

Method A prioritizes stability and reliability, whereas Method B prioritizes sensitivity to small movements. Therefore, the two methods have different strengths rather than one method being universally better.

Analysis

For a dynamic scene, small movements can be important. Method A may provide smooth and reliable motion estimation but could fail to detect subtle movements. Method B can capture those small movements, but its noisy vectors may produce unstable tracking.

For example, in human-motion analysis, small movements of the hands or face may be important, making sensitivity valuable. However, in vehicle tracking, unstable vectors caused by noise could result in incorrect motion estimates.

Application

If the application requires precise detection of small movements, Method B is more suitable, provided that a noise-reduction or smoothing stage is added. If the main requirement is stable and reliable tracking of larger motion, Method A is preferable.

For a general dynamic-scene tracking system, an improved approach would be to use Method B with noise filtering and vector smoothing so that its fine-motion capability is retained without excessive instability.

Conclusion

Neither method is best for every situation. Method A is better for stable tracking, while Method B is better for detecting fine motion. For dynamic scenes where small movements matter, Method B with appropriate noise reduction would provide the better solution.

4. **Background Modeling Strategy (GMM) A GMM-based system shows:**

- **High false positives in dynamic background**
- **Missed detections in static background**

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

Given Data

The GMM-based system shows:

- High false positives in dynamic backgrounds
- Missed detections in static backgrounds

Data Interpretation

A Gaussian Mixture Model (GMM) is being used to model the background and separate moving objects from it. The results indicate that the model does not perform consistently in different background conditions.

In a dynamic background, elements such as moving trees, water, or changing shadows may be incorrectly classified as foreground, producing false positives.

In a static background, the system may fail to detect some actual objects because their appearance can be similar to the learned background or because the background model is not updated appropriately.

Analysis

The two problems show that using GMM alone is not sufficient for reliable foreground detection. A dynamic background requires the system to distinguish genuine object motion from natural background movement. A static environment requires accurate separation of newly appearing objects from the established background.

Therefore, the system needs additional processing instead of relying only on the GMM output.

Proposed Improvement

A better system can use:

GMM Background Modeling → Noise Removal → Morphological Processing → Object Validation → Final Detection

Morphological operations can remove small false regions and connect fragmented foreground regions. Adaptive background updating can also help the model respond to gradual environmental changes.

Conclusion

GMM alone is not sufficient for reliable detection in both dynamic and static backgrounds. Combining GMM with morphological filtering, adaptive background updating, and object validation can reduce false positives and missed detections.

5. **Motion Estimation vs Computational Cost Two methods:**

- **Method A: low error (5%), high computation**
- **Method B: moderate error (10%), low computation**

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

Motion Estimation vs Computational Cost

Given Data

Method	Error	Computational Cost
Method A	5%	High
Method B	10%	Low

Data Interpretation

Method A has the lower error rate of 5%, meaning it provides more accurate motion estimation. However, it requires high computational resources. Method B has twice the error at 10%, but its computational requirement is low.

Therefore, the choice depends on whether accuracy or processing speed is the main requirement.

Analysis

For real-time applications, the system must process frames quickly. A high-computation method may introduce delays even though it gives better accuracy. Therefore, Method B may be more practical when immediate response is important.

For offline applications, processing time is less critical because results do not need to be produced immediately. In this situation, the lower error of Method A becomes more valuable.

Application

Application	Preferred Method	Reason
Real-time video tracking	Method B	Low computation and faster processing
Offline motion analysis	Method A	Lower error and higher accuracy

If Method B is selected for real-time use, additional optimization or filtering can be considered to reduce its 10% error without significantly increasing computational cost.

Conclusion

Method B is more suitable for real-time applications because its low computational cost supports faster processing. Method A is better for offline applications where accuracy is more important than processing speed. The final choice should therefore depend on the application's timing and accuracy requirements.